# Berkeley Event Value Twin

*Kite Festival prototype — complete value, not only permit cost. Select scenario on Controls sheet.*

| Active scenario | Central |
|---|---|

| City net fiscal impact | Net direct local spending | Total economic output |
|---|---|---|
| $45,748 | $723,594 | $1,027,503 |

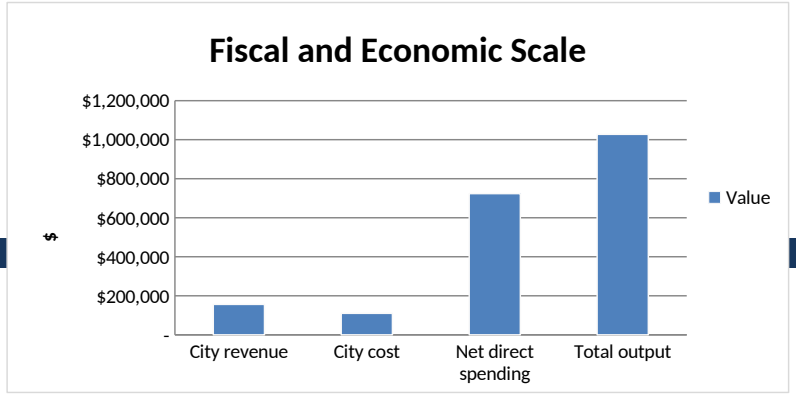
| Realized attendance | Vehicle arrivals | Transportation emissions |
|---|---|---|
| 32,200 | 8,318 | 93.3 |

| City cost recovery | Public-value score | Parking surplus / (shortfall) |
|---|---|---|
| 141.6% | 70.2 | (3,989) |

## Interpretation

The active scenario is estimated to cover incremental City cost while also producing local economic and public-value effects.

| Metric | Value |
|---|---|
| City revenue | $155,748 |
| City cost | $110,000 |
| Net direct spending | $723,594 |
| Total output | $1,027,503 |




## Uncertainty summary

| | |
|---|---|
| Expected fiscal impact | $32,811 |
| P10 fiscal impact | $15,878 |
| Median fiscal impact | $32,883 |
| P90 fiscal impact | $49,138 |
| Probability of break-even | 99.8% |

# Berkeley Event Value Twin — Controls

*Blue cells are editable. Values in v1 are illustrative placeholders until Berkeley records and survey data are loaded.*

## Scenario selection

| | | |
|---|---|---|
| Active scenario | Central | *Choose Conservative, Central, or High* |

| Assumption | Conservative | Central | High | Unit | Status / source |
|---|---|---|---|---|---|
| Total attendance | 25,000 | 35,000 | 45,000 | people | *Illustrative; replace with event counts* |
| Berkeley resident share | 45.0% | 35.0% | 25.0% | % | *Survey / ZIP code* |
| Festival-attributed share: nonresidents | 65.0% | 78.0% | 88.0% | % | *Intercept survey* |
| Festival-attributed share: residents | 5.0% | 12.0% | 20.0% | % | *Intercept survey* |
| Avg Berkeley spend: resident attendee | $18 | $25 | $34 | $/person | *Intercept survey* |
| Avg Berkeley spend: nonresident day visitor | $35 | $52 | $72 | $/person | *Intercept survey* |
| Overnight visitor share of nonresidents | 2.0% | 5.0% | 9.0% | % | *Intercept survey* |
| Avg incremental lodging spend | $145 | $185 | $240 | $/room-night | *Survey / hotel data* |
| People per hotel room | 1.8 | 2.0 | 2.2 | people | *Survey* |
| Local capture of attendee spending | 72.0% | 82.0% | 90.0% | % | *Location of expenditure* |
| Economic output multiplier | 1.25x | 1.42x | 1.58x | x | *IMPLAN; RIMS II cross-check* |
| Value-added share of total output | 48.0% | 54.0% | 60.0% | % | *IMPLAN sector results* |
| Labor-income share of total output | 28.0% | 34.0% | 40.0% | % | *IMPLAN sector results* |
| Sales-tax effective rate | 0.8% | 0.9% | 1.0% | % of taxable sales | *Replace with City fiscal estimate* |
| Taxable share of incremental spending | 65.0% | 72.0% | 80.0% | % | *Spending category mix* |
| City service cost | $135,000 | $110,000 | $90,000 | $ | *Illustrative; reconstruct marginal cost* |
| Permit and organizer fees paid to City | $30,000 | $45,000 | $60,000 | $ | *Illustrative* |
| Parking price per vehicle | $10 | $15 | $20 | $ | *Scenario policy* |
| Parking revenue retained by City | 75.0% | 85.0% | 90.0% | % | *Contract / fund allocation* |
| Auto mode share | 72.0% | 62.0% | 50.0% | % | *Mobility survey* |
| Average vehicle occupancy | 2.1 | 2.4 | 2.7 | people/vehicle | *Mobility survey* |
| Average round-trip miles | 28 | 34 | 40 | miles/vehicle | *Origin model* |
| Transit mode share | 10.0% | 15.0% | 22.0% | % | *Mobility survey* |
| Bike/walk mode share | 12.0% | 17.0% | 23.0% | % | *Mobility survey* |
| Vehicle emissions factor | 0.0% | 0.0% | 0.0% | metric tons CO2e/mi | *Fleet assumption* |
| Waste per attendee | 75.0% | 60.0% | 45.0% | lb/person | *Event waste audit* |
| Waste diversion rate | 45.0% | 60.0% | 75.0% | % | *Event operations* |
| Weather realization factor | 78.0% | 92.0% | 1.0 | % of planned atten | *Weather risk* |
| Displaced local spending | $60,000 | $35,000 | $15,000 | $ | *Nearby business / visitor survey* |

## Active scenario values (formula-driven)

| Assumption | Active value | Unit |
|---|---|---|
| Total attendance | 35,000 | people |
| Berkeley resident share | 35.0% | % |
| Festival-attributed share: nonresidents | 78.0% | % |
| Festival-attributed share: residents | 12.0% | % |
| Avg Berkeley spend: resident attendee | $25 | $/person |
| Avg Berkeley spend: nonresident day visitor | $52 | $/person |
| Overnight visitor share of nonresidents | 5.0% | % |
| Avg incremental lodging spend | $185 | $/room-night |
| People per hotel room | 2.0 | people |
| Local capture of attendee spending | 82.0% | % |
| Economic output multiplier | 1.42x | x |
| Value-added share of total output | 54.0% | % |
| Labor-income share of total output | 34.0% | % |
| Sales-tax effective rate | 0.9% | % of taxable sales |
| Taxable share of incremental spending | 72.0% | % |
| City service cost | $110,000 | $ |
| Permit and organizer fees paid to City | $45,000 | $ |
| Parking price per vehicle | $15 | $ |
| Parking revenue retained by City | 85.0% | % |
| Auto mode share | 62.0% | % |
| Average vehicle occupancy | 2.4 | people/vehicle |
| Average round-trip miles | 34 | miles/vehicle |
| Transit mode share | 15.0% | % |
| Bike/walk mode share | 17.0% | % |
| Vehicle emissions factor | 0.0% | metric tons CO2e/mile |
| Waste per attendee | 60.0% | lb/person |
| Waste diversion rate | 60.0% | % |
| Weather realization factor | 92.0% | % of planned attendance |
| Displaced local spending | $35,000 | $ |

# Visitor Economics

*Incremental spending only: attribution, local capture, lodging, displacement, and multiplier effects.*

| Attendance segmentation | | |
|---|---|---|
| Realized attendance | 32,200 | *Planned attendance × weather realization* |
| Berkeley residents | 11,270 | |
| Non-Berkeley attendees | 20,930 | |
| Incremental resident attendees | 1,352 | |
| Incremental nonresident attendees | 16,325 | |
| Overnight visitors | 1,047 | |

| Direct incremental spending | |
|---|---|
| Resident attendee spending | $27,724 |
| Nonresident day-visitor spending | $651,492 |
| Incremental lodging spending | $79,377 |
| Gross direct incremental spending | $758,594 |
| Less displaced local spending | ($35,000) |
| **Net direct incremental spending** | $723,594 |

| Regional economic effects | |
|---|---|
| Total economic output | $1,027,503 |
| Value added | $554,852 |
| Labor income | $349,351 |
| Indirect + induced output | $303,909 |

| Fiscal bridge | |
|---|---|
| Estimated taxable incremental sales | $520,987 |
| Estimated City sales-tax revenue | $4,689 |

# City Fiscal Impact

*Separates true incremental cash cost from allocated accounting cost and compares City revenues.*

| City revenues | | |
|---|---|---|
| Permit and organizer fees | $45,000 | |
| Estimated sales-tax revenue | $4,689 | |
| Parking vehicles | 8,318 | |
| Gross parking revenue | $124,775 | |
| City-retained parking revenue | $106,059 | |
| **Total City revenue** | $155,748 | |

| City costs | | |
|---|---|---|
| Incremental City service cost | $110,000 | |
| Police / traffic | $38,500 | 35.0% |
| Fire / EMS | $13,200 | 12.0% |
| Public Works / sanitation | $19,800 | 18.0% |
| Parks / Marina staff | $19,800 | 18.0% |
| Temporary facilities / contracts | $13,200 | 12.0% |
| Administration / permitting | $5,500 | 5.0% |
| **Allocation check** | $110,000 | 100.0% |

| Net fiscal result | | |
|---|---|---|
| Net City fiscal impact | **$45,748** | *Positive = surplus; negative = subsidy* |
| City subsidy per attendee | - | |
| City cost recovery ratio | 141.6% | |

# Mobility & Emissions

*Event travel, parking pressure, transportation emissions, and solid waste.*

## Travel demand

| | | |
|---|---|---|
| Realized attendance | 32,200 | |
| Auto travelers | 19,964 | |
| Average vehicle occupancy | 2.4 | |
| Vehicle arrivals | 8,318 | |
| Transit travelers | 4,830 | |
| Bike/walk travelers | 5,474 | |
| Other / unallocated travelers | 1,932 | |

## Parking and roadway indicators

| | | | |
|---|---|---|---|
| Usable event parking supply | 2000 | spaces | *Illustrative; replace with current inventory* |
| Peak parking demand | 5,989 | *Illustrative 72% peak overlap* | |
| Parking surplus / (shortfall) | (3,989) | | |
| Parking utilization | 299.5% | | |

## Emissions and waste

| | | |
|---|---|---|
| Vehicle miles traveled | 282,823 | |
| Transportation emissions | 93.3 | *metric tons CO2e* |
| Total event waste | 9.7 | *short tons* |
| Landfilled waste | 3.9 | *short tons* |
| Diverted waste | 5.8 | *short tons* |

# Public Value Scorecard

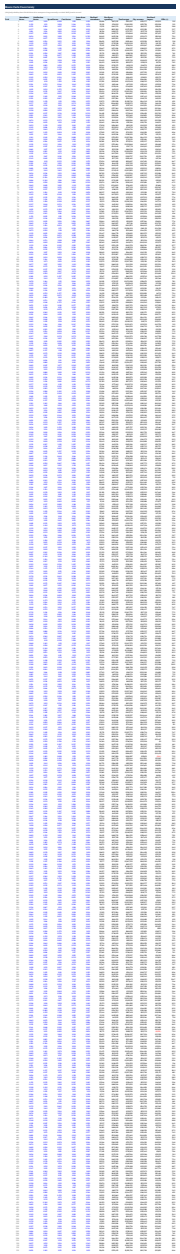
*Nonmarket benefits remain visible as measured outcomes instead of being forced into unsupported dollar values.*

## Weighted scorecard

| Outcome | Weight | Score (0–100) | Weighted score | Evidence status | Measurement instrument |
|---|---|---|---|---|---|
| Free family recreation | 18.0% | 75 | 13.5 | Illustrative | Attendee survey / observation |
| Accessibility and inclusion | 14.0% | 65 | 9.1 | Illustrative | Accessibility audit + survey |
| Youth engagement | 12.0% | 70 | 8.4 | Illustrative | Program counts + survey |
| Berkeley identity / visibility | 14.0% | 80 | 11.2 | Illustrative | Media analytics + resident survey |
| Waterfront activation | 12.0% | 78 | 9.4 | Illustrative | Footfall / dwell time |
| Local nonprofit/vendor exposure | 10.0% | 68 | 6.8 | Illustrative | Vendor follow-up survey |
| Resident satisfaction | 12.0% | 62 | 7.4 | Illustrative | Representative resident survey |
| Environmental stewardship | 8.0% | 55 | 4.4 | Illustrative | Waste / habitat / travel indicators |
| **Total / composite score** | 100.0% | | 70.2 | | |

## Distributional checks

| | | |
|---|---|---|
| Share of attendees from Berkeley | 35.0% | Target / survey |
| Share using non-auto modes | 32.0% | Mobility survey |
| City subsidy per attendee | - | Fiscal model |
| Free admission | 100.0% | Event design |
| Parking shortfall | 3989.2 | Mobility model |





## Evidence Register

*Tracks whether each material input is observed, modeled, stipulated, or still missing.*

| Domain | Input / evidence | Status | Confidence | As-of date | Source / custodian | URL or record request | Model owner | Update rule |
|---|---|---|---|---|---|---|---|---|
| Attendance | 2017–2019 hourly/daily attendance | Missing | Low | | Organizer / City special events | | | Replace scenario range |
| Fiscal | Police and traffic marginal cost | Missing | Low | | Berkeley Police / Finance | https://berkeleyca.gov/your-government/public-records | | Reconstruct overtime and contracts |
| Fiscal | Fire/EMS marginal cost | Missing | Low | | Berkeley Fire / Finance | https://berkeleyca.gov/your-government/public-records | | Reconstruct event-specific cost |
| Fiscal | Marina/Parks staffing and sanitation | Missing | Low | | Parks, Recreation & Waterfront | https://berkeleyca.gov/your-government/public-records | | Separate fixed from incremental |
| Economy | Creative Economy impact methodology | Available | High | 2025 | City of Berkeley | https://berkeleyca.gov/sites/default/files/documents/Berkeley-Cre | | Use consistent geography and outputs |
| Economy | Attendee spending by origin | Missing | Low | | Future intercept survey | https://www.americansforthearts.org/by-program/reports-and-data | | Survey every pilot event |
| Economy | Regional multiplier set | Modeled | Medium | | IMPLAN + BEA RIMS II | https://www.bea.gov/resources/methodologies/RIMSII-user-guide | | Refresh to analysis year |
| Mobility | Waterfront parking/TDM framework | Available | High | 2025 | City of Berkeley | https://berkeleyca.gov/sites/default/files/documents/Berkeley-WTP | | Update when plan changes |
| Mobility | Actual event mode share and occupancy | Missing | Low | | Future mobility survey | https://highways.dot.gov/media/87306 | | Survey and counts |
| Environment | Event waste and diversion | Missing | Low | | Event operator / Zero Waste | | | Weigh or audit loads |
| Community | Resident and attendee public-value measure | Missing | Low | | Representative survey | https://www.americansforthearts.org/by-program/reports-and-data | | Pre/post or event survey |
| Geospatial | Open Data assets and boundaries | Available | High | | City of Berkeley | https://berkeleyca.gov/your-government/public-records/open-data- | | Refresh annually |

## Data Acquisition Plan

*Records and measurements needed to replace illustrative assumptions with Berkeley-specific evidence.*

| Priority | Request / measurement | Custodian | Historical period | Format | Reason | Minimum fields | Status |
|---|---|---|---|---|---|---|---|
| 1 | Event invoices, permits, and organizer payment | Finance / PRW | 2017–2019 | CSV/PDF | Reconstruct City fiscal le | date, account, description, amount, payer/payee | Not started |
| 1 | Event-specific staff hours and overtime | Police, Fire, Public Works, PRW | 2017–2019 | CSV | Estimate marginal servic | department, employee class, regular/overtime hours, rate | Not started |
| 1 | Attendance estimates and methodology | Organizer / City | 2017–2019 | CSV/PDF | Anchor historical event s | date, hour, gate/zone, method, estimate | Not started |
| 1 | Vendor, sponsor, and organizer budgets | Organizer | 2017–2019 | CSV/PDF | Map event money flows | category, local/nonlocal, amount | Not started |
| 2 | Parking counts, traffic plans, transit operations | Transportation / AC Transit | 2017–2019 | CSV/PDF | Calibrate mobility model | facility, time, capacity, occupancy, service | Not started |
| 2 | Nearby sales indicators or anonymized tax aggr | Finance / economic development | Comparison weekends | Aggregated CSV | Estimate net local effects | district, sector, period, taxable sales | Not started |
| 2 | Waste tonnage and cleanup records | Public Works / operator | 2017–2019 | CSV/PDF | Environmental impact | stream, weight, cost, date | Not started |
| 3 | Media reach and public sentiment | Organizer / City | 2017–2019 | CSV/PDF | Civic visibility and value | channel, reach, geography, sentiment | Not started |
| Pilot | Attendee intercept survey | Independent survey team | Future event | Survey data | Incrementality and spen | ZIP, party size, trip purpose, spending, mode | Design needed |
| Pilot | Resident comparison survey | Independent survey team | Before/after pilot | Survey data | Distribution and public v | residency, use, support, burden, WTP optional | Design needed |

# Model Guide

*Version 1.0 — working prototype for a Berkeley Kite Festival / civic-event counterfactual assessment.*

## Purpose

This workbook compares a festival scenario with a no-event counterfactual across City finances, local economics, mobility, emissions, and public value. It is a decision model—not evidence that the event should or should not return.

## How to use

1. Select Conservative, Central, or High on Controls!B5. 2. Replace blue illustrative assumptions with observed Berkeley data. 3. Review Dashboard and component sheets. 4. Use Evidence Register and Data Requests to document gaps. 5. Rebuild the fixed Monte Carlo draws if uncertainty structures change materially.

## Interpretation

Economic output is not City revenue. City fiscal impact is not total public value. Gross attendee spending is reduced by attribution, local capture, and displacement before multipliers are applied.

## Color conventions

Blue text = editable input; green text = linked from another sheet; black = formula; gray = static label; orange = missing/review item; purple = scenario control; teal = key result.

## Model limitations

All event-specific numerical assumptions in v1 are illustrative unless replaced and documented. The model does not yet include anonymized card transactions, mobile-location data, a full traffic microsimulation, habitat valuation, or a statistically representative resident survey.

## Recommended next step

Reconstruct 2017–2019 City and organizer accounts, then design a one-day pilot with an intercept survey, parking/traffic counts, waste measurement, and pre-specified adjustment procedures.